Performance Requirements/Specifications

Barillas Peak Lookout Tower Restoration

SANTA FE NATIONAL FOREST

05/04/2026

This document is intended to establish the requirements & expectations for the functionality, operability, sustainability, and overall serviceability of the Barillas Peak & Encino Vista Lookout Towers on the Santa Fe National Forest. Additionally included are the performance requirements for the contractor, their products and their execution of contract specified work.

NOTE: Each lookout tower has a unique scope of work. The contract Statement of Work in conjunction with the bid schedule distinguishes the required work, systems and/or items for each site individually. This document establishes the minimum requirements of said systems, work and/or items.

Barillas Peak/Encino Vista Lookout Tower: Performance Requirements/Specifications

1.0 SCOPE: The contractor shall provide services for lookout tower restoration work as noted in the Statement of Work. This Scope section consolidates all technical, construction, design, demolition, safety, communications, utilities, power, and grounding requirements described throughout this SOW. The contractor shall provide a full turnkey solution including: Design Phases 0-100%; tower structural upgrades; canopy/cab fabrication and installation; heating; mechanized lift installation; solar PV system with battery bank and inverter; DC/AC power distribution; gray water and plumbing systems; CMU bathroom/storage building; lightning protection and grounding per NFPA 70 and NFPA 780; communications equipment including ANT150F2 antenna installation and cable routing; site civil work including trenching, fencing, graveling and reseeding; demolition of existing cab and debris disposal; safety compliance including OSHA and USFS standards; protection of property; environmental compliance; project management and reporting; and all testing including concrete, RF sweeps, and grounding OHM testing. All requirements in Sections 4.0–17.0 remain fully applicable. The contractor shall be required to obtain construction bonding and requisite insurance sufficient to support the individual construction of towers, as appropriate.

2.0 SOW FOR DESIGN PHASE:

Phase 1: Project Initiation and Existing Conditions Assessment (0%)

- **End-User Requirements Meeting:** Conduct a kickoff meeting with USFS staff (the end-users) to finalize operational needs, desired layout, and specific requirements for the habitable canopy space.
    - **Site Visit and Data Collection:** Conduct a site reconnaissance to assess the extent of the fire damage, confirm existing dimensions, and evaluate site logistics (access, geotechnical conditions, etc.).
    - **Existing Structure Documentation:** Document the current condition and material properties of the existing steel tower structure to inform the structural analysis.
    - **Utility Planning:** Investigate and document the existing septic tank to confirm its size, condition, and determine the optimal connection points. Also, identify optimal locations and constraints for the solar array and water system components.
    - **Load Determination:** Calculate the structural loads associated with the proposed new replacement canopy, of-grid solar equipment, and water storage/delivery systems (davit).

Phase 2: Preliminary Design (30% Submission)

This phase establishes the foundational design concepts and spatial relationships, ensuring alignment with USFS operational requirements before proceeding to detailed engineering.

1. **Structural Scheme:**
    - **Existing Tower Analysis:** Perform a preliminary structural analysis of the existing steel tower to confirm its capacity to support the new replacement canopy building and associated systems (solar, water loads, etc.).
    - **Canopy Design Concept:** Develop schematic plans, elevations, and a structural concept for the replacement of canopy building.
2. **Of-Grid Systems Design:**
    - **Solar System:** Determine energy consumption needs and develop a preliminary of-grid solar photovoltaic (PV) array size and battery storage concept.
    - **Water/Waste Plan:** Develop a preliminary concept for clean water storage, gray water collection, and the connection plan to the existing septic tank, based on anticipated flow requirements. ***(Note: Design of a new tank or a drain field***

is explicitly excluded from this scope.)

3. **Davit System:**
    - Specify the type and capacity of the davit arm and winch mechanism required to lift clean water storage containers up to the canopy level.

4. **Site Plan:**
    - Develop a 30% draft Site Plan showing the location of the tower, existing septic tank, water delivery zones, and solar array footprint.

Phase 3: Intermediate Design Review (60% Submission)

This phase serves as a critical gate check to refine the 30% concepts, finalize major engineering decisions, and
secure USFS alignment before proceeding to full documentation.

1. **Preliminary Sizing and Modeling:** Complete initial, model-based sizing and preliminary component selection for the structural members, solar array, and battery storage.
2. **Interdisciplinary Coordination:** Finalize critical interfaces, ensuring the davit mounting loads and the solar array supports are fully integrated into the canopy structural frame and tower connections.
3. **60% Document Production:** Issue partially coordinated plans, elevations, and preliminary engineering calculations for formal USFS review and feedback.

Phase 4: Design Development (90% Submission)

This phase incorporates the 60% review comments and finalizes all construction details, leading to a near-complete set of documents **ready for construction**.

1. **Detailed Structural Design:**
    - **Tower:** Final structural analysis report validating the capacity of the existing tower. Detail any necessary strengthening/remediation to the existing tower structure due to fire damage or load increases.
    - **Canopy:** Complete structural design and detailing for the replacement of canopy building, including foundation, framing, and attachment details to the existing tower.
2. **Utility and Plumbing Design:**
    - **Water System:** Final design of the clean water storage, distribution lines, and gray water drainage within the canopy.
    - **Wastewater:** Document the **existing septic tank** dimensions and detail the final plumbing connection design (in-flow piping) from the replacement canopy building to the existing tank.
    - **Davit System:** Final specification and integration details for the davit and winch, including
      mounting loads for structural tie-in.
3. **Solar Design:**
    - Finalize the design, panel layout, mounting details, wiring schematics, and equipment
      specifications for the of-grid solar power system.
4. **90% Construction Documents:**
    - Produce fully coordinated drawings (Architectural, Structural, Civil, Electrical/Solar,
      Plumbing/Wastewater) and draft technical specifications for all components.

Phase 5: Final Design and Construction Documents (100% Submission)

This final phase includes necessary revisions, quality control checks, and the final production of a complete
set of construction documents.

- **QA/QC and Review:** Incorporate all USFS comments from the 90% review phase.
- **Final Production:** Complete all outstanding details, schedules, and cross-discipline coordination.
- **Stamping and Sealing:** Professional Engineer (PE) stamping and sealing of all required Construction Documents (Structural, Civil, etc.) **ready for construction and permitting**.

5.0 SITE VISIT:

The contractor may need to perform site visits prior to commencing construction of lookout cab and related supporting utilities. Should site visits be required, the contractor shall provide the government radio technician with the date, time, and meeting place for the site visit. The government radio technician can advise the contractor what types of vehicles may be necessary for traversing the area.

6.0 TOWER: Barillas Lookout Tower

All specifications shall meet as a minimum, ANSI TIA/EIA 222 Rev H. "Structural Standards for Antenna Supporting Structures, Antennas and Small Wind Turbine Support Structures" based on:

Height of Tower: Existing 54' USFS Lookout Tower
Wind & Ice Load Calculations: Based on ANSI EIA/TIA 222-H.
Equipment Load Calculations: Based on proposed loading.
Risk Category (Class of Structure): Structure Class category as appropriate.
Exposure Categories: Exposure category as appropriate.
Topographic Categories: Topo category as appropriate.

Tower shall have Solid Core, Angle Iron, or sealed tubular legs and are to be hot dipped galvanized. Tower design shall support wind loading of lookout cab, wrap around platform, all antennas, mounts, and coaxial cables as proposed.

Prebuild build and As-built designs shall be provided showing the following.

Tower diagram with a table identifying antenna types,
heights. Rolling sphere diagram including all air terminals
and antennas

Grounding system includes ground rods, ground test well, bus bars, and all connection points i.e., tower legs, etc.

Include a digital set of P.E. Stamped tower drawings stamped with the seal of the state where the tower is being built.

7.0 TOWER: Encino Vista Lookout Tower

All specifications shall meet as a minimum, ANSI TIA/EIA 222 Rev H. "Structural Standards for Antenna Supporting Structures, Antennas and Small Wind Turbine Support Structures" based on:

Height of Tower: Existing 54' 6" USFS Lookout Tower
Wind & Ice Load Calculations: Based on ANSI EIA/TIA 222-H.
Equipment Load Calculations: Based on proposed loading.

Risk Category (Class of Structure): Structure Class category as appropriate.
Exposure Categories: Exposure category as appropriate.
Topographic Categories: Topo category as appropriate.

Tower shall have Solid Core, Angle Iron, or sealed tubular legs and are to be hot dipped galvanized. Tower design shall support wind loading of lookout cab, wrap around platform, all antennas, mounts, and coaxial cables as proposed.

Prebuild build and As-built designs shall be provided showing the following.

Tower diagram with a table identifying antenna types,
heights. Rolling sphere diagram including all air terminals
and antennas
Grounding system includes ground rods, ground test well, bus bars, and all connection points i.e., tower legs, etc.

Include a digital set of P.E. Stamped tower drawings stamped with the seal of the state where the tower is being built.

8.0 SOW FOR CANOPY, LIFT, UTILITIES, POWER, COMMUNICATIONS, LP/GROUNDING, CIVIL:

(*Existing Staircase Tower with Lookout Cab)*

- **New Canopy/Cab — Fabrication & Installation**
  - **Design, fabricate, and install** a new steel canopy/cab sized for the tower top, including floor, wall, and roof structural members as outlined in the ROM *(hot-dip galvanized, with wall siding).*
  - **Handrail/guardrail assembly** *(galvanized)* around catwalk per ROM.
  - **Bottom-of-stairs access gate** *(if directed by USFS*
  - **Hatch:** Provide new safe/lockable hatch *(CL100-type)* with counter-weight or improved assist mechanism; hazard mitigation for heavy leaf doors.
  - **Door & Windows:** Install new operable/fixed windows *(12 ea. per ROM concept)* designed for high winds and weather; weather-stripped door preferably on north/east exposure per end-user input; include high-visibility screens.
  - **Metal roll-down shutters** for winterization *(rolling type preferred)* for all windows/required openings. *(preferred over side-to-side);* ensure weather-tightness and locking; shutters must not block views when stowed.
  - **Insulation & interior paneling:** Install insulation and FRP paneling inside; frame out walls with wood studs as needed for finishes.
  - **Interior finishes/flooring:** Durable, non-carpet flooring; functional workplace residence configuration.

- **Interior Outfitting, Furnishings & Fire Finder**
  - Supply/Install fire finder, fire-finder cabinet *(wood),* shelving, two twin beds with storage *(foam mattresses, no metal springs),* kitchen cabinetry and sink/countertop (no backsplash), table, chairs *(all wood),* bookcase, blinds *(no strings),* and weather station *(e.g., Ambient WS-2902).* Metal appliances to be bonded to LP system where applicable.
  - Supply/Install appliances: electric refrigerator *(compact with freezer),* small microwave, two-burner 1,800W hot plate, *(small electric water heater-currently not priced).*
  - Provide adequate AC outlets for electronics and operations per layout; LED lighting *(dimmable preferred; warm color).*

- **Mechanized Lift / Davit**

- **Provide and install a mechanized lift system** *(winch/pulley with telescopic/rotating arm or approved davit)* for safe hoisting of supplies/water to canopy; integrate into structural design; include load rating plates and safety features.

- **Utilities — Plumbing & Gray Water**
  - **Install gray-water drainage from canopy to bathroom**; trenching and piping as required.
  - **Install cistern for gray-water collection** and connect to existing septic per design; no new tank/drain field design in scope.
  - **New bathroom concrete building** *(CMU block 8×8×9 approx.)* including toilet, sink rough-in, storage shelving/capabilities; materials testing, compaction testing, and rebar inspection required. Shower excluded per USFS direction.

- **Electrical Power, Solar, and Distribution**
  - **Install solar array with DC power backup** and **DC-AC conversion**; provide battery/inverter system per design. Current concept: ~5.2 kW PV with 48 VDC battery bank and inverter solution; final sizing & mounting per A&E.
  - **Run power from battery cabinet/AC converter then up tower**; provide branch circuits/outlets/lighting as designed; integrate ground heating *(electric baseboard),* refrigerator, microwave, stove top/hot plate, outlets, fire finder power, and other misc. loads.
  - **Backup generator**: Install sustainable backup system.
  - **Heating:** Provide electric baseboard heating with thermostatic control; no AC unless directed; design for proper ventilation per window/operable openings strategy and high-elevation conditions.

- **Communications & Technology**

  - **Antennas:** If specific antennas are stated in the task order, no antenna substitutions are allowed without acceptance of the COR. All antennas shall be identified and a diagram with manufacturer type, ownership, height, and leg/face on the tower shall be provided upon completion.
  - **Antenna:** Supply and install **one (1) ANT150F2** VHF antenna mounted above canopy roofline, with LDF4-50 coax run, entry ports *(1×1 4"),* external/internal ground bus bars, Polyphaser VHF50HN within 24" of entry, proper labeling, and sweep test reports *(PDF)* for USFS approval before crew leaves site.
  - **Provide antenna mounting structure/hardware** allowing maintenance without tower-mast climbing where feasible *(e.g., slider bracket solution)* and position to avoid window blind spots.
  - **Antenna and coaxial cable installation:** shall be connected to the single point ground system shall be supplied with the tower, all materials shall be installed in accordance with and in compliance with National Electrical Code *(NFPA 70)* and National Fire Protection Association *(NFPA 780)* for Installation of Communications Equipment.
  - **Entry panels:** supply and install (2) entry panels, one on the exterior wall of the building and the second on interior wall. Minimum of (1) 4" entry ports shall be installed at building entrance and shall include all necessary weatherproof boots and cushions to accommodate the required number of antenna cables. All unused holes shall be sealed with appropriate product plugs (silicone or foam is not acceptable). Must allow for the minimum expansion of (2) additional antenna cables.
  - **Cable Runs:** Heliax cable specified in the Tower Request Form appropriate for the run length *(Minimum of LDF4-50)* shall be used for all runs. Vendor must supply

documentation of cable specifications. All cables shall be secured to tower using appropriate hangers. Cables shall be installed on the cable ladder in an orderly fashion, with the highest antenna lines to be toward the center of the ladder. All cables shall be grounded at the antenna, at the bottom of the tower and at the building entrance as stated above. Coaxial cables shall be labeled at the point where they leave the tower and at the inside termination. This labeling shall include the antenna make and model, frequency range, and location on the tower. The interior labeling may be labelling printer tape, and the outside labeling shall be stamped metal, laminated cardstock, or similar weatherproof method. All cables shall be terminated with N Male connectors (or as needed for direct attachment to antenna and Polyphaser protector). Cable shield ground attachment points and antenna connections shall be weatherproofed to industry standards, using mastic and tape or cold shrink protection, if mastic is used; a courtesy wrap (Single wrap of electrical covering both end of the connector) tape shall be installed on antenna connections. Coaxial cables shall be secured to the standoff using an appropriate hanger *(zip ties or metal banding will not be acceptable.)* Coaxial line lightning/surge protection device, similar to Polyphaser VHF50HN will be installed near to the internal Master Ground Bus and grounded to the Master Ground Bus. MGB location to be determined at time of installation.

- **Lightning Protection (LP) & Grounding**
  - **Lightning protection:** Air terminal will be installed no less than 6 feet above the tallest antenna and in a way the does not affect the radiation pattern of any antennas in a negative fashion. An isolated ground cable shall connect the air terminal to the ground system following guidelines called out in NFPA 780.
  - **Install LP system** meeting **NFPA 780** with the following baseline configuration:
    - (4) corner 15' × 2" Sch. 80 galvanized pipe extensions, each supporting 48" × 5/8" air terminals, with 2/0 tinned copper stranded 28T downleads to tower legs; clamps/ brackets/ connectors per ROM; include additional air terminal for existing telecom skid as listed.
  - The Forest Service uses the **National Electrical Code (NFPA 70) and National Fire Protection Association (NFPA 780)**. It is recommended that all personnel responsible for the installation, testing, and maintenance of Forest Service grounding and lightning protection systems reference the NEC, and NFPA 780 standards for more detail on the subjects contained in this document.
  - **Fall of Potential Method:** The contractor shall perform ground resistivity testing using the Fall of potential (also known as the 3-point fall method) is the most recognized method for measuring the
    resistance to earth of a grounding system, or ground system performance. Pre and Post tests shall be documented and supplied to the FS POC.
  - **Grounding Performance Objectives:** Conductivity-to-ground is the primary goal for an effective grounding system. The acceptable maximum results shall be ten (10) ohms of resistivity. Post installation, contractor shall perform a final system test for compliance and report the results.
  - **Grounding system** to include a ground ring, (14) 3/4"×10' copper-clad ground rods, #2 tinned solid conductors, exothermic welds, bus bars (cab & cabinet), inspection well(s), and ground enhancement material (GEM/bentonite) per ROM bill of materials.
  - **OHM testing** (ground resistance) and 3rd-party certification by TEDS; provide photos of all below-grade grounding and obtain USFS civil inspector & COR sign-off prior to burial.
  - **Properly ground** the tower, canopy, solar A-frame structure, equipment cabinets, and mini telecom skid per design.
  - **Ground Dispersion System:** The contractor shall design and install a ground system

around the base of the existing tower, new solar array, existing telecom skid w/ tower, ground cow tracker skid, propane tank (if kept), according to the NEC, NFPA 780, and TIA-222 current addition. The resistance shall conform to TIA-222H standards of ten (10) ohms or less. Vendor to demonstrate and provide certification of measurement using the Standard Three Point/Fall of Potential method. A proposed design layout will be provided with initial bid with final change to be included in the final design. All ground field connections to the tower shall be exothermic welds using #2 solid copper cable or larger. The vendor shall connect to all metal structures within 15' of the nearest approach of the ground field conductors. This includes, but not limited to, buildings, site fencing, propane tanks, generator, exterior ground bar, etc. These connections to be via exothermic weld when possible or a non-reversible ground lug connection otherwise. If ground lugs are used, they are to be of the appropriate finish to prevent corrosion due to galvanic action.

- **Copper Ground Bar:** Copper ground bar, minimum 2" x 12" x ¼" shall be installed beneath the buildings entry panel(s), interior and exterior. All ground bars shall be connected to the grounding conductors of the ground field using #2 solid copper cable or larger. The below grade grounding connections shall be made via exothermic weld. Each coaxial cable shall be grounded, using the appropriately sized ground strap, to the tower at the antenna, to the tower ground bar at the tower base and to the external ground bus bar at the building entrance. All ground bars shall be tinned copper. Attachment of the tower ground bar to the below grade ground may be with tinned non-reversible compression lugs and tinned #2 solid copper cable. Ground attachment points on the tower may make use of existing bolts or may use beam clamps of galvanized or stainless-steel construction. An internal Master Ground Bus and external Ground Bus shall be installed within 24 inches of the building entry port, be installed below entry panel to allow downhill flow of transient energy to the ground system. It shall be connected to the single point ground system using methods described in this section.

- **Site/Civil Work**
  - **Trenching** for water pipe and DC power runs; backfill/compaction.
  - **New handrail** and any new fencing on the way up to meet USFS specifications; repair fencing where damaged.
  - **Rock/gravel** *(washed rock)* the access road, turnaround, and parking per ROM pricing options; reseed non-rocked disturbed areas with native grass seed *(weed-free)*.
  - Maintain **heliport** if directed by USFS.
  - **Access to Site:** Public roads to site shall be accessible with travel available late spring through late fall. At the site, it shall be the contractor's responsibility to provide access to the construction site for their heavy equipment. If no access road is provided, it is the responsibility of the vendor to return any and all areas used to access the construction site to their original conditions. It shall be the vendor's responsibility to repair any damage to access road by project equipment and return it to its previous condition.
  - **Excess Earth:** All excess soils will be dispersed according to the local Forest Service units' regulations.
  - **Garbage:** The construction site shall be rid of garbage daily to avoid trash being spread by weather or animals. All garbage will be disposed of per Forest Service instructions.

  - **Revegetation:** revegetate extensive cut and fill slopes with native vegetation as soon as possible after construction. All revegetation must have prior written approval of the authorized officer. Revegetate with grass mix approved by local USFS office.

8.0 DEMOLITION:

- The removal of existing lookout tower cab will be done in a way that no damage will occur to remaining structures/equipment.
- Any damage caused by the removal of the structure will be the responsibility of the contractor to repair.
- Remove existing flagpole with associated foundation *(if required).*
- Disposal of all demolition debris is the responsibility of the contractor and shall be discarded in accordance with state, federal environmental regulations as applicable.

9.0 SAFETY PLAN

- Provide detailed description of a safety plan for review.
- Vendors must comply will all Occupational Safety & Health Standards (OSHA) Safety Standards for climbing and tower installation.
    - Information may be obtained at www.osha.gov or you can contact OSHA 800-321-6742.
- Vendor must provide fencing around the immediate installation of the communication tower and foundation when not working at site.

10.0 Acceptance Criteria

- **Design acceptance:** USFS approval at each phase gate and at 100% CD.
- **LP/Grounding:** Meets NFPA 780 design intent; documented continuity; ground resistance values per TEDS design report; USFS civil inspector sign-off before backfill.
- **Structure/Canopy:** Installed per stamped drawings/specs; windows/doors/shutters operate correctly; handrails/fencing secure; hatch safe/lockable.
- **Utilities:** Gray-water drains as designed; fixtures operate; solar/inverter/battery supports stated loads; power distribution functional; heaters operate within design parameters.
- **Site/Civil:** Rocking and reseeding completed; disturbed areas stabilized; access gate (if installed) functional; fence repairs complete.
- **Concrete Tests:** Perform the following tests prior to placement of concrete, Slump, Temperature, Air Content, and cast a minimum of one set of test cylinders (four cylinders per set) for every truckload of concrete placed.
- **Sweep Tests:** Antenna systems shall be tested to ensure they meet manufacturer specifications for frequency, return loss, and distance to fault. These tests shall be saved and provided to the USFS as proof of functionality of each antenna system. **RF systems:** Sweep test traces meet manufacturer specs for frequency/return loss/DTF; USFS approval prior to demobilization.

Contractor Obligations

1.1 Contractor General Responsibilities

1.1.1 The Contractor shall submit a management plan within 60 days of award that describes the intended approach that meets the specifications of this SOW. The management plan will consist of the following documents at a minimum:

a. **Project Management Plan**: Contractor shall provide a project plan in Microsoft Excel consisting of a Work Breakdown Structure (WBS) and associated project schedule identifying each task and subtask and delivery timelines for each. The project schedule will additionally identify minor

and major milestone completion dates for each communications tower to be designed, manufactured, and installed. Contractor shall assign a project manager to provide complete oversight of the project. The Project Manager will be responsible for maintaining and updating the project schedule and all other duties associated with the project. The Contractor's Project Manager will be the direct interface with the Forest Service COR and must be reasonably available to the USFS during the prime hours of work 0800 - 1630 Mountain Time, Monday to Friday. An alternate Project Manager POC must be provided in case they are unreachable or on leave.

b. **Identification of Contractor Employees.** All contract personnel attending meetings, answering government telephones, and working in other situations where their contractor status is not obvious to third parties are required to identify themselves as such to avoid creating an impression in the minds of members of the public that they are government officials. They must also ensure that all documents or reports produced by contractors are suitably marked as contractor products or that contractor participation is appropriately disclosed. Contractor personnel will not be required to obtain and wear contractor badges in the performance of this service.

c. **Subcontractor/Supplier Management Plan**: Contractor shall inform the USFS COR of all subcontractors or suppliers used during the duration of the project. The Contractor shall provide a Delegation of Authority letter to the COR within 10 days of ARO. Contractor will be held responsible for the performance of subcontractors and suppliers and properly manage and provide adequate oversight to protect the respective forest. The COR shall be notified immediately if a Government employee or Contractor deems a subcontractor or supplier as unfit, unsafe, unethical and should be replaced and/or removed immediately as to protect the interests of the Government and its citizens.

d. **Risk Management Plan**: Contractor will submit a risk management plan within 30 days of contract award. The risk management plan will detail anticipated risk areas and provide options for mitigating such risks and will include at minimum a Risk Register and Risk assignment matrix.

e. **Fire Plan**: Prior to initiating work on the contract area, during the closed fire season period, a fire prevention and suppression plan will usually be prepared. The Contractor and the Forest Service will jointly prepare this plan. Such plan shall include a detailed list of men and equipment at the Contractor's disposal for implementing the plan. The fire plan shall also specify additional measures and/or special requirements, such as Hoot Owl restrictions, necessary during periods of critical fire weather conditions.

f. **QA/QC Plan**: Contractor will submit a Quality Assurance Plan within 30 days of contract award. The QA/QC plan will detail all QA/QC processes to include quality audits and describe the process for remedy and non-recurrence of defective services and equipment in the event of quality violations. Contractor will perform quality control audits at a

minimum of two intervals per shelter: at hand-off of shelter to the contractor from USFS at the staging area and at completion of installation. The QA plan is not to be confused with the QASP.

g. **Acceptance Plan**: The Contractor will submit a Quality Inspection Checklist with each tower installation for acceptance and signature by the Forest Service POC. The Quality Inspection Checklist will validate compliance to all requirements set forth in this SOW as well as those provided in established manufacturer guidelines.

h. **Post Award Conference/Periodic Progress Meetings.** The Contractor agrees to attend any post award conference convened by the contracting activity or contract administration office in accordance with Federal Acquisition Regulation Subpart 42.5. The Contracting Officer, Contracting Officers Representative, and other government personnel, as appropriate, may meet periodically with the contractor to review the contractor's performance. At these meetings the contracting officer will apprise the contractor of how the government views the contractor's performance and the contractor will apprise the Government of problems, if any, being experienced. Appropriate action shall be taken to resolve outstanding issues. These meetings shall be at no additional cost to the government.

1.1.2 Provide weekly and monthly reports of daily activities to the Forest Service POC.

1.1.3 Submit IPP invoices to include breakdown of work done with costs and pictures.

1.1.4 Ensure that all deliverables for all tasks under this contract are complete, accurate and delivered according to established schedules and are provided to the Government for review, comment, and acceptance 60 days prior to installation or as soon as possible.

1.1.5 Ensure the appropriate oversight and quality control of all activities included in this SOW follow applicable Federal, State and local agency laws, regulations, manufacturer specifications, policy and ensure all USFS regulations and policies are followed while on site.

1.2 Protection of Property

1.2.1 Underground utilities

Locate underground utilities prior to excavation. Contractor is responsible for utility location. If utilities are damaged during excavation, contact the Contracting Officer immediately and take necessary measures to prevent interruption of service (if live). Damaged utilities will be repaired at the Contractor’s expense.

1.2.2 Traffic

Conduct all operations to ensure minimum interference and do not close or obstruct roads and other adjacent occupied or used facilities without permission from the COR and Local Forest POC.

1.2.3 Environmental Compliance

The contractor needs to comply with the Environmental Management System (EMS) provided. by the forest service. The EMS provides controls to manage impacts on National Forests and Grasslands. These controls include NEPA analysis, Best Management Practices (BMP), and resource monitoring to ensure controls are in place and working as intended.

Environmental Objectives:

a. Invasive Weeds – Reduce invasive weed introduction or spread through improved and more effective control and management. New infestations of noxious weeds of concern to the Forest Service and identified by either Contractor or Forest Service, on the Project Area or on the haul route, will be promptly reported to the other party. Contractor and Forest Service will agree on treatment methods to reduce or stop the spread of noxious weeds when new infestations are found. A current list of noxious weeds of concern to the Forest Service is available at each Forest Service office.

b. Protection of existing trees, vegetation, improvements and sites. Damaged resources must be reported to the CO prior to attempts at remediation. Solvents, oils and other materials shall be disposed of in containers and removed from the site. Any contaminated soils shall be removed from National Forest land and replaced with topsoil.

c. Travel Management – Reduce the existing road systems' contributed sediment and impairment of water quality and aquatic habitats and improve security for grizzly bears or elk through the designation of roads, trails, and areas for motor vehicle use.

1.3 Security Requirements:

1.3.1 PHYSICAL Security: The contractor shall be responsible for safeguarding all government property provided for contractor use. At the close of each work period, government facilities, equipment, and materials shall be secured.

1.3.2 Key Control:

The Contractor shall establish and implement methods of making sure all keys/key cards issued to the Contractor by the Government are not lost or misplaced and are not used by unauthorized persons. The Contractor shall prohibit the use of Government issued keys by any persons other than the Contractor's employees. The Contractor shall prohibit the opening of locked areas by Contractor employees to permit entrance of persons other than Contractor employees engaged in the performance of assigned work in those areas, or personnel authorized entrance by the Contracting Officer. In the event keys, other than master keys, are lost or duplicated, the Contractor shall, upon direction of the Contracting Officer, re-key or replace the affected lock or locks; however, the Government, at its option, may

replace the affected lock or locks or perform re-keying. When the replacement of locks or re-keying is performed by the Government, the total cost of re-keying or the replacement of the lock or locks shall be deducted from the monthly payment due the Contractor. In the event a master key is lost or duplicated, all locks and keys for that system shall be replaced by the Government and the total cost deducted from the monthly payment due the Contractor. No keys issued to the Contractor by the Government shall be duplicated. The Contractor shall develop procedures covering key control that shall be included in the Quality Control Plan. Such procedures shall include turn-in of any issued keys by personnel who no longer require access to locked areas. The Contractor shall immediately report any occurrences of lost or duplicate keys/key cards to the Contracting Officer.

1.4 Completion Criteria: The contractor will have fulfilled its obligations under this acquisition when the following occurs:

a. All relevant project documents have been collated and provided to the US Forest Service as a final closeout package. To be accepted the closeout package must include the following: PE stamped tower and foundation drawings, ground system drawings, rolling sphere drawing, as-built drawings, geotechnical report, concrete mix, concrete testing reports, antenna sweeps tests, and fully signed certificate of completion.

b. All services have been received and accepted.

c. The period of performance ends or the Forest Service terminates the provisions of the contract.

1.5 Miscellaneous:

1.5.1 Work Site: The contractor shall keep the work site free from accumulation of waste materials and rubbish resulting from the work. Waste material shall be disposed of legally off project and National Forest Land. Before final acceptance of the project by the Government, the contractor shall remove from Government land all waste materials and rubbish resulting from this work.

1.5.2 Barricades, Warning Signs, and Other Devices: The Contractor shall provide, erect and maintain warning signs in advance to any place on the project where operations may interfere with the use of the road or trail by traffic and at all intermediate points where the new work crosses or coincides with an existing road or trail to protect both the work and public. All road barricades, warning signs, lights, temporary signals, flagmen and pilot car operators and equipment, and other projective devices, except for special devices, shall conform to Part VI of the Manual on Uniform Traffic Control Devices for Streets and Highways, published by the Federal Highway Administration and applicable safety codes.

1.5.3 Hours of Operation/Place of Performance: The contractor is responsible for providing services as pre-coordinated with the local contact except Federal holidays or when the Government facility is closed due to local or national emergencies, administrative closings, or similar Government directed facility

closings. The Contractor must at all times maintain an adequate work force for the uninterrupted performance of all tasks defined within this statement of work when the Government facility is not closed for the above reasons. Request for work on the weekends shall be coordinated and approved by the Local Forest and coordinated with COR within a reasonable amount of time to secure all approvals as required.

1.5.4 Recognized Holidays:

Government personnel and potentially facilities may not be available during federal holidays. Federal holidays can be found via OPM's website at https://www.opm.gov/policy-data-oversight/pay-leave/federal-holidays/#url=Overview.

Fire Control 1.6

(a) Contractor's Responsibility for Contractor-Caused Fires. The Contractor, whether or not directed by the Forest Service, shall immediately extinguish, without expense to the Government, all fires on or in the vicinity of the project which are caused by Contractor's employees, whether set directly or indirectly as a result of Contractor operations. The Contractor may be held liable for all damages and costs of additional labor, subsistence, equipment, supplies, and transportation resulting from fires set or caused by the Contractor's employees or resulting from contract operations.

(b) Other Fires. For the purpose of fighting forest fires on or in the vicinity of the project which are not caused by the Contractor or the Contractor's employees, the Contractor when requested by the Contracting Officer shall place the employees and equipment temporarily at the disposal of the Forest Service. Payment for such services will be made by the Government at not less than the current rate for fire-fighting services established by the Forest Service in the area concerned.

Any employees and equipment furnished will be relieved from firefighting as soon as the Forest Service finds that it is practicable to employ other labor and equipment adequate for the protection of the area.

An equitable adjustment in contract time may be made for this period.

(c) Fire Protection Requirements - Fire Plan. At all times during closed fire season period, as specified by State law, the Contractor shall comply with each of the following provisions to the extent applicable to the Contractor's operation under the contract.

(1) Fire Tools. The Contractor will provide for each employee in the contract area at least one approved hand tool of a type appropriate in the contract area, such as shovel, Pulaski, or ax. Tools required and furnished under (2) and (4) below, shall count toward fulfillment of the above requirement. Where additional tools, beyond those required under (2) and (4) below, are to be provided,

the Contractor shall seal such tools in one or more boxes painted red and marked "Tools for Fire Only." All tools required herein shall be kept sharp and in good serviceable condition and maintained at locations to be designated by the Forest Service.

(2) *Fire Extinguishers and Tools on Mobile or Stationary Equipment.* Each unit of powered equipment used in connection with this contract, including automobiles, trucks, tractors, etc., shall be equipped with serviceable tools and fire extinguishers as follows:

One - fire extinguisher, dry chemical type of not less than 2-1/2-pound capacity with 4 BC or higher rating. One - shovel, round point #0 lady or equal.

One - ax, 2 pounds or over, 26-inch minimum length, or one Pulaski.

One - water container (at least 1-gallon capacity), not required with stationary equipment.

(3) Spark Arresters. Each internal combustion engine shall be provided with a spark arrester or spark arresting device approved by the Forest Service. Exceptions where the Forest Service may approve mufflers, or other equipment in lieu of spark arresters qualified and rated under Forest Service Standard 5100-1a are: (a) small multilocation engines, such as chain saws, shall meet Society of Automotive Engineers J335b standards; (b) passenger-carrying vehicles and light trucks may have baffle-type mufflers with tail pipe; (c) heavy-duty trucks may have a vertical stack exhaust system and muffler, provided the exhaust stack extends above the cab of the vehicle. An exhaust-driven turbocharger is considered to be a satisfactory spark arrester. Internal combustion engine exhaust systems, arresters, and other devices must be properly installed and maintained.

(4) Power saws. For each power saw used in connection with this contract, the following will be provided:

One - shovel, round point #0 lady or equal. Shovel must be immediately available for use.

One - Fire extinguisher, containing not less than 8 ounces of extinguisher fluid, or a dry chemical powder-type of not less than 1-pound capacity. The extinguisher must be immediately accessible to the saw operator at all times.

Any fueling or refueling of a power saw shall only be done in an area which has first been cleared or is free of all material capable of carrying fire; power saw shall be moved at least 10 feet from place of fueling before starting.

(5) Blasting. Fuse or prima cord shall not be used unless authorized in writing by the COR with special precautions stated.

(6) Smoking. Smoking shall not be permitted within the contract area except on surfaced or dirt roads, at landings, within closed vehicles, in camps, or at other posted places, and shall never be allowed while working or traveling on foot.

(7) Storage of Petroleum and Other Highly Flammable Products. Gasoline, oil, grease, or other highly flammable material will be stored either in a separate building used exclusively for such storage, or at a site where all combustible debris and vegetation is cleared away within a radius of 25 feet. Fire extinguishers and/or sand barrels may be required at such locations specified by the Forest Service when unusually hazardous conditions exist.

(8) Debris Burning and Warming Fires. Burning permits will be required for all debris burning fires. Lunch and warming fires may be allowed in fireproofed areas during periods of low fire danger if and as specified in the fire plan. Such fires must not be left burning unattended.

(9) Precautions for Stoves. Stovepipes on all temporary buildings, trailers, and tents using wood burning stoves, will be equipped with roof jacks and serviceable spark arresters of mesh with openings no larger than 5/8 inch.

All stovepipes, inside and out, will not be closer than 2 feet from any wood or other flammable material or 1 foot if the combustible material is protected by a metal or asbestos shield.

(10) Welding. Welding or use of cutting torches will be permitted only in areas that have been cleared or are free of all material capable of carrying fire. Flammable debris and vegetation must be removed from within a minimum of 10 feet radius of all welding and cutting torch operations. A shovel and a 5-gallon standard backpack water container (filled) with handpump attached, shall be immediately available for use in the event of a fire start.

(11) *Fire Plan.* Prior to initiating work on the contract area, during the closed fire season period, a fire prevention and suppression plan will usually be prepared. Such plan shall include a detailed list of men and equipment at the Contractor's disposal for implementing the plan. The fire plan shall also specify additional measures and/or special requirements, such as Hoot Owl restrictions, necessary during periods of critical fire weather conditions.

When Hoot Owl restrictions are invoked, the Forest Service may curtail or shut down all or portions of a contractor's operations. The following requirements may be imposed:

(i) All high fire risk operations could be terminated at 1300 local time.

(ii) All burning could be stopped, including debris burning fires.

(iii) Patrolman may be required for a period of 2 hours after high fire risk operations and 1 hour after end of work shift. The Contractor shall provide the patrolman unless the contract is for equipment rental, in which case the Government will provide the patrolman.

(12) *Pump and Trailer. <u>If applicable to the Task Order, the Contracting Officer will state so in the Task Order Request.</u>* The Contractor shall provide at a location satisfactory to the Contracting Officer, a serviceable truck or trailer, equipped with a firefighting tanker unit to be kept ready for instant use for suppressing forest fires. The unit shall consist of a tank of not less than 100-gallon capacity upon which shall be mounted a

live hose reel or live hose basket with 250 feet of at least 3/4-inch I.D. heavy-duty rubber hose; a portable or power takeoff pump with discharge capacity of at least 10 gallons per minute at 150 P.S.I. pressure. Gear type pumps shall be provided with a bypass or pressure relief valve so that the hose nozzle may be shut while the pump is operating. Each tanker unit shall have a hose nozzle of the shut-off type, adjustable for straight stream, spray or fog, at least 12 feet of 1-inch suction hose with an intake screen, and additional 250 feet of 3/4 –inch heavy-duty rubber hose or 1-inch cotton jacket rubber-lined or linen hose to be carried on the unit for use as needed. Tools, adapters, accessories and fuel necessary to operate the pump and truck or trailer shall be provided. If a trailer is used, a serviceable vehicle with proper trailer tow hitch shall be located at a point satisfactory to the Contracting Officer. Where water is available, a supply sufficient for rapidly filling the water tank shall be provided at one or more accessible points along or adjacent to the main truck roads.

(13) *Burning.* Before starting any open burning, the Contractor shall comply with the following:

(i) Submit a burning plan, subject to approval by the COR/Inspector, designed to minimize the impact on air quality and to lessen any fire damage.

(ii) Obtain a burning permit from the District Ranger.

(iii) Use weather forecasts as far as possible in scheduling burning for more favorable dispersal of smoke.

(iv) Fuel shall be reasonably free of dirt and piled to facilitate rapid and complete combustion. Piles shall be tended during burning operations to maintain combustion and to eliminate unnecessary smudges. Piles shall be patrolled during off-work hours to avoid wildfires.

(v) Unless fire season controls dictate otherwise, burning shall be scheduled, as far as practicable, for the most favorable conditions during the burning period.

(vi) Under certain atmospheric conditions, smoke accumulations may become excessive, and, in these cases, the Regional Forester may, at his/her discretion, prohibit all open burning for definite periods within specific areas of the National Forest.

Objective/Performance Measurements: 2

2.1 The contractor will accurately perform the duties assigned to them within these error rates:

a) Project Management:

1. Safety and security protocols are adhered to 100% of the time with no at-fault reportable incidents.
2. Ensure compliance with all USFS, local, State and Federal laws and Regulations 100% of the time, with no reported violations. Comply with all local forest requirements with respect to fire, hours of work, noxious weed precautions and any additional requirements as conditions dictate.
3. Provide Risk Management Plan to COR within 30 days of award.
4. Provide Project Management plan to the COR within 60 days of award 100% of the time.
5. Provide subcontractor Delegation of Authority letter within 15 days of ARO 98% of the time.
6. Bi-weekly summary reports within 7 days of completion with no more than 2 reports 5 business days late in a 12-month period.
7. Monthly summary report of contract within 5 days after the end of the month, no more than 2 reports 5 business days late in a 12-month period.
8. Provide final metrics and a Quality Assurance Plan (QAP) within 30 days of contract award 98% of the time.
9. Provide a detailed drawings of the tower to the COR no later than 30 days after GeoTech report is received. 98% of the time.
10. Provide a project timeline/schedule within 30 days after the award is received. 100% of the time.
11. Provide a proposed delivery and installation schedule coordinated with each forest within 60 days after Government Furnished Information (GFI) is fully receipted by the contractor 100% of the time.
12. Provide a Detailed installation plan for all sites within no later than 30 days prior to installation. This will include Site specific list of tasks, times, and locations for every installation. (Who, What, Where)

13. Provide a finalized project timeline/schedule for all sites within 7 days after COR Approval of proposed installation schedule 98% of the time.

14. Submission of accurate Invoices to COR without rejection 98% of the time.

15. The contractor shall provide skilled (manufacture trained as required) personnel together with the supervision, management and administrative services necessary to successfully meet the requirements of this contract. 100% of the time.

b) Tower Delivery and Installation:

1. Fire prevention plan will be coordinated with the Forest prior to initiating work at the site. The Fire prevention plan will be submitted to the COR prior to initiating work 100% of the time.

2. Provide a detailed Air Lift safety plan associated with the Air Lift installation to include maps, load plans, and number of flights within 60 days prior to the event 100% of the time.

3. Deliver specified tower to correct location 100% of the time.

4. Install specified tower according to manufacturer's specifications 100% of the time, unless otherwise specified by the COR.

5. Install specified tower according to the latest version of the COR approved schedule 98% of the time.

6. Provide accurate, complete, detailed report of each Airlift event to include copies of flight plans (provided before each event) for each lift, and copies of flight time log and fuel logs for each event within 15 business days 100% of the time. Must be signed by USFS POC or Engineer on site.

7. Provide an acceptance plan for each tower installation. The contractor will submit a Quality inspection checklist with each tower installation for acceptance and signature by the Forest Service POC in order to validate compliance. The Quality inspection checklist is required for closeout.

c) Existing Tower Removal *(if required)*

1. Remove tower and foundation to 18” below grade specified by the COR 100% of the time.

2. Disposal or relocation of existing tower to location specified by the government 100% of the time.

3.2 Quality Assurance Surveillance Plan: The QASP is put in place to provide Government surveillance and oversight of the contractor's efforts in meeting the performance standards required by the contract. The QASP is not part of the contract, nor is it intended to duplicate the Contractor's quality control plan. The Government may provide the Contractor with an informational copy of the QASP to support the Contractor's efforts in developing a quality control plan that will interrelate with the QASP.

3.1 Methods of QA Surveillance: The surveillance methods listed below shall be used in the administration of this QASP. The specific surveillance method for each performance standard and acceptable quality level is listed in the "Monitoring Method" column of the SOW.

a. **COR Review –** Indicates that the COR will be responsible for monitoring the Contractor's performance in meeting specific performance standards/acceptable quality levels.

1. In the case of a written deliverable, COR/COR Designee Review will consist of up to 100% inspection of the specific deliverable and supporting documentation.

2. In the case of day-to-day support activities, COR/COR Designee Review will consist of periodic inspection or random monitoring.

b. Forest POC Review – Indicates that the Forest POC will be responsible for monitoring the Contractor's performance in meeting specific performance standards/acceptable quality levels.

3. In the case of a written deliverable, POC Review will consist of up to 100% inspection of the specific deliverable and supporting documentation.

4. In the case of day-to-day support activities, POC Designee Review will consist of inspection of each delivery, installation or removal event.

c. Customer Feedback – Customer feedback may be obtained either from the results of formal customer satisfaction surveys or from customer complaints received by the Forest Service. Customer complaints, to be considered valid, must set forth clearly and in writing the detailed nature of the complaint, and must be signed and forwarded to the COR, who shall assess the validity of the complaint and retain it in a documentation file. The COR shall ensure the tabulated results of all customer satisfaction surveys are kept on file.

d. **Monitoring:** The results of all monitoring activity shall be documented on the Surveillance Activity Summary Form. The COR and COR

Designee shall submit copies of the Summary Form to the Contracting Officer at the conclusion of contract performance, and whenever contractor performance fails to meet acceptable quality levels.

3.2 Documentation: The COR and/or COR Designee will, in addition to providing documentation to the Contracting Officer, maintain a complete quality assurance file. The file will contain copies of all reports, evaluations, recommendations, surveillance checklists, and any actions related to the Government's performance of the quality assurance surveillance function. All such records will be retained for the life of the contract and forwarded to the Contracting Officer upon completion.